Siddharth Mishra

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About

Computer Science undergraduate with hands-on experience across software development and AI engineering, building production-grade ML pipelines, RAG-based GenAI applications, and full-stack web systems using modern development practices.

Education

B.Tech in Computer Science and Technology | CGPA: 8.98/10

Maharaja Agrasen Institute of Technology (2023 - 2027)

- Relevant Coursework: Data Structures & Algorithms, Machine Learning, Database Management, Computer Networks, Artificial Intelligence

Skills

- **Languages:** C++, Java, Python.
- **AI & Machine Learning:** Numpy, Pandas, Feature Engineering, Predictive Modeling, Deep Learning, Natural Language Processing, LLM Application, Langchain, RAG, LangGraph.
- **Web Development:** HTML, CSS, JavaScript, React.js, Next.js, Node.js, MongoDB.
- **Tools & Platforms:** GitHub, Figma, VS Code, Vercel, Jupyter Notebook, Docker, Streamlit, AWS, LangSmith.

Projects

AGRIINSIGHT - Crop Yield Intelligence System

Python, Scikit-learn, Streamlit, Supervised Learning

- Trained and optimized a Random Forest regression model on **340,000+ agricultural records**, achieving **$R^2 = 0.80$** for district-level crop yield prediction through feature engineering and hyperparameter tuning.
- Built an **end-to-end ML pipeline** using preprocessing, target encoding, and one-hot encoding to generate accurate crop yield forecasts across multiple regions and crop types.
- Developed an interactive **Streamlit dashboard** for yield prediction, historical crop trend analysis, district-level insights, and model explainability using feature importance techniques.

CAREERLENS - AI Resume Analysis Platform

Python, FastAPI, LangChain, LangGraph, Gemini API, Next.js

- Built an agentic AI application that leverages **Gemini LLM** to perform intelligent resume analysis, identify technical competencies, and generate personalized, context-aware evaluations.
- Designed a **4-stage LangGraph** orchestration pipeline for document processing, prompt engineering, structured reasoning, and response synthesis, improving modularity and extensibility.
- Developed a **FastAPI backend** integrating AI inference, user authentication, and persistent storage for scalable API-driven services.

YT GPT - RAG Based YouTube Chatbot

Python, GenAI, FAISS, Streamlit

- Built an **8-stage** Retrieval-Augmented Generation (RAG) pipeline covering transcript ingestion, preprocessing, chunking, embeddings, vector indexing, retrieval, prompt engineering, and Gemini-powered answer generation.
- Implemented semantic search using **FAISS vector databases** with multilingual transcript support, duplicate chunk removal, source citations with timestamps, and conversational memory for context-aware Q&A.
- Developed and deployed a **Streamlit-based chatbot** integrating YouTube Transcript API, Sentence Transformers, and Gemini API with robust handling for transcript failures, retrieval inconsistencies, and API errors.

Leadership and Certifications

- Served as Vice President, Rotaract Club MAIMS & MAIT - led event planning and outreach initiatives
- Artificial Intelligence, Samsung Innovation Campus (Dec 2025 – Mar 2026)
- Complete Data Science, Machine Learning, Deep Learning & NLP Bootcamp - Udemy, 99 hours (Feb 2026)